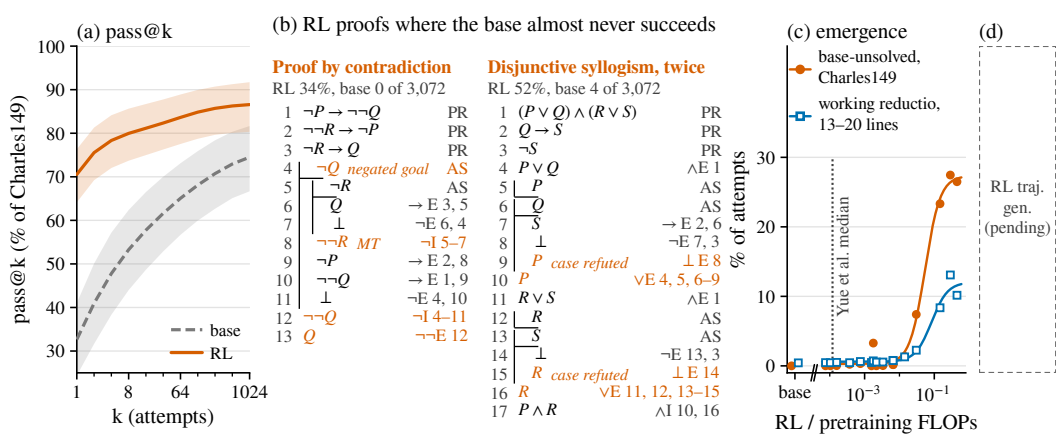
(a) pass@k
pass@k (% of Charles149)
k (attempts)
base
RL
(b) RL proofs where the base almost never succeeds
Proof by contradiction
RL 34%, base 0 of 3,072
Disjunctive syllogism, twice
RL 52%, base 4 of 3,072
negated goal
MT
case refuted
(c) emergence
base-unsolved, Charles149
working reductio, 13–20 lines
Yue et al. median
% of attempts
RL / pretraining FLOPs
(d)
RL traj. gen. (pending)